

Course Notes for MATH 375: Probability Theory

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Lecture 1

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1 Lecture 1: Sets and Counting

1.1 Union and intersection

The **union** $A \cup B$ contains every element that belongs to A , to B , or to both:

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

The **intersection** $A \cap B$ contains the elements shared by both sets:

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

For example, if

$$A = \{1, 2, 5, 7\}, \quad B = \{2, 3, 6, 8\},$$

then

$$A \cup B = \{1, 2, 3, 5, 6, 7, 8\}, \quad A \cap B = \{2\}.$$

1.2 Special sets and set properties

The **universal set** U contains all elements under discussion. The **empty set** $\emptyset$ contains no elements.

Some useful identities are

$$\begin{aligned} A \cup B &= B \cup A, & A \cap B &= B \cap A, \\ A \cup \emptyset &= A, & A \cap \emptyset &= \emptyset, \\ A \cup U &= U, & A \cap U &= A. \end{aligned}$$

The distributive laws are

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

and

$$(A \cap B) \cup C = (A \cup C) \cap (B \cup C).$$

1.3 Complements and De Morgan's laws

The **complement** A^c contains the elements of U that are not in A :

$$A^c = \{x \in U : x \notin A\}.$$

De Morgan's laws are

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

1.4 Cardinality and inclusion-exclusion

For a finite set A , its **cardinality** $|A|$ is its number of elements. To count the elements in a union, use

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

We subtract $|A \cap B|$ because adding $|A|$ and $|B|$ counts the shared elements twice.

For three sets,

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| \\ &\quad + |A \cap B \cap C|. \end{aligned}$$

1.5 The multiplication principle

If a task has several successive choices, multiply the number of options at each step.

Bit strings. Each position in a bit string can be either 0 or 1. Therefore, the number of bit strings of length n is

$$\underbrace{2 \cdot 2 \cdots 2}_{n \text{ positions}} = 2^n.$$

License plates. Suppose a plate has four digits followed by three letters, and repetition is allowed. There are

$$10^4 \cdot 26^3$$

possible plates.

If the four digits must be distinct and the three letters must be distinct, the count becomes

$$(10 \cdot 9 \cdot 8 \cdot 7)(26 \cdot 25 \cdot 24) = 78,624,000.$$

Area codes. If an area code has three digits and its first digit cannot be 0 or 1, while the other digits have no restrictions, there are

$$8 \cdot 10 \cdot 10 = 800$$

possible area codes.

1.6 Choosing positions: combinations

How many bit strings of length n have exactly one 0? Choose its position. There are n choices.

How many have exactly two 0s? Choose two of the n positions:

$$\binom{n}{2} = \frac{n!}{2!(n-2)!} = \frac{n(n-1)}{2}.$$

For instance, with $n = 5$,

$$\binom{5}{2} = 10.$$

Decision cue: When identical special objects must occupy certain positions, choose the positions. You do not need to arrange the remaining identical objects separately.

1.7 Permutations

A **permutation** of n distinct objects is an ordering of all n objects. The number of such orderings is

$$n(n-1)(n-2) \cdots 2 \cdot 1 = n!.$$

Decision cue: If changing the order creates a different outcome, think about permutations. If you only need to select which positions or objects are involved, think about combinations.